

The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

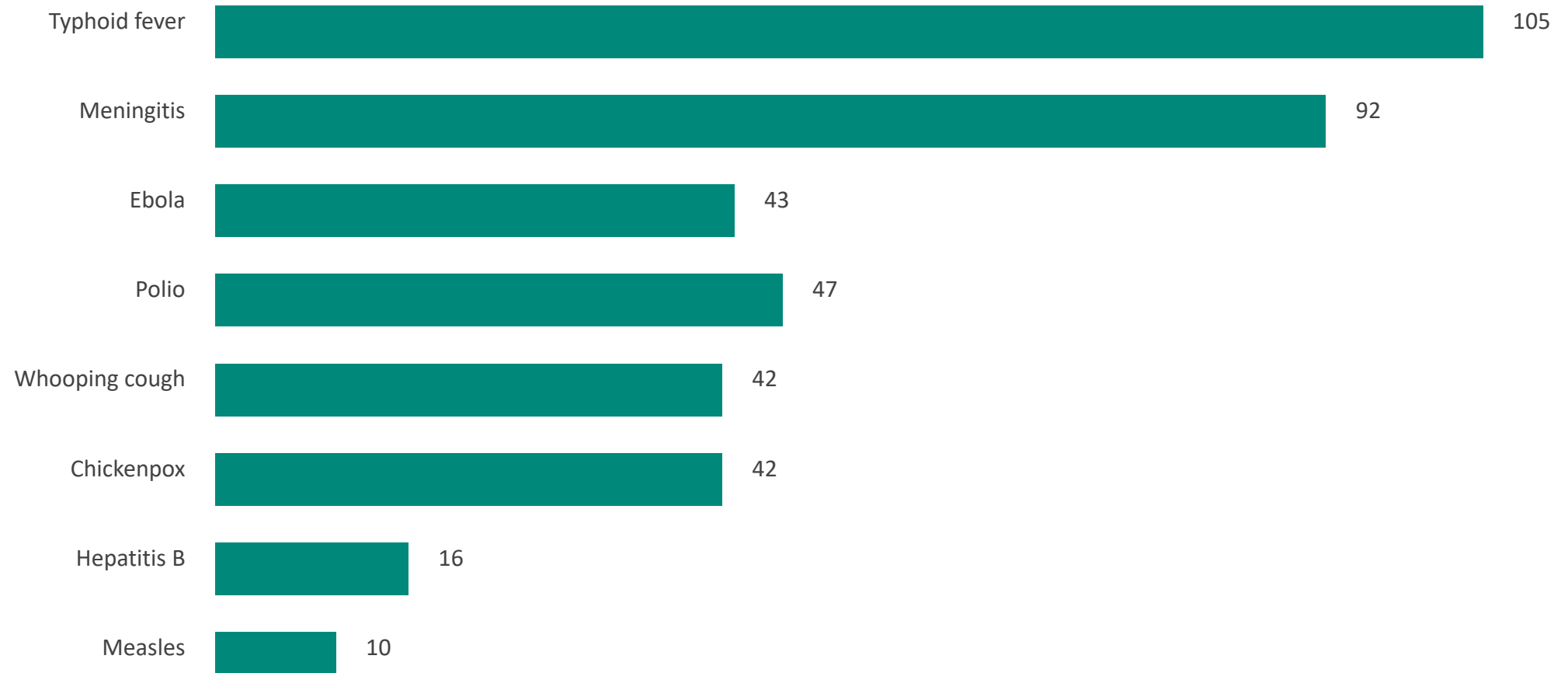
A Data-Driven Overview of Vaccine Impact | 1796 – 2020

Data source: Our World in Data — Vaccination

For public health professionals and policy analysts

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Years from pathogen identification to vaccine licensure



COVID-19: mRNA technology and genomic tools compressed development from decades to under 1 year

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

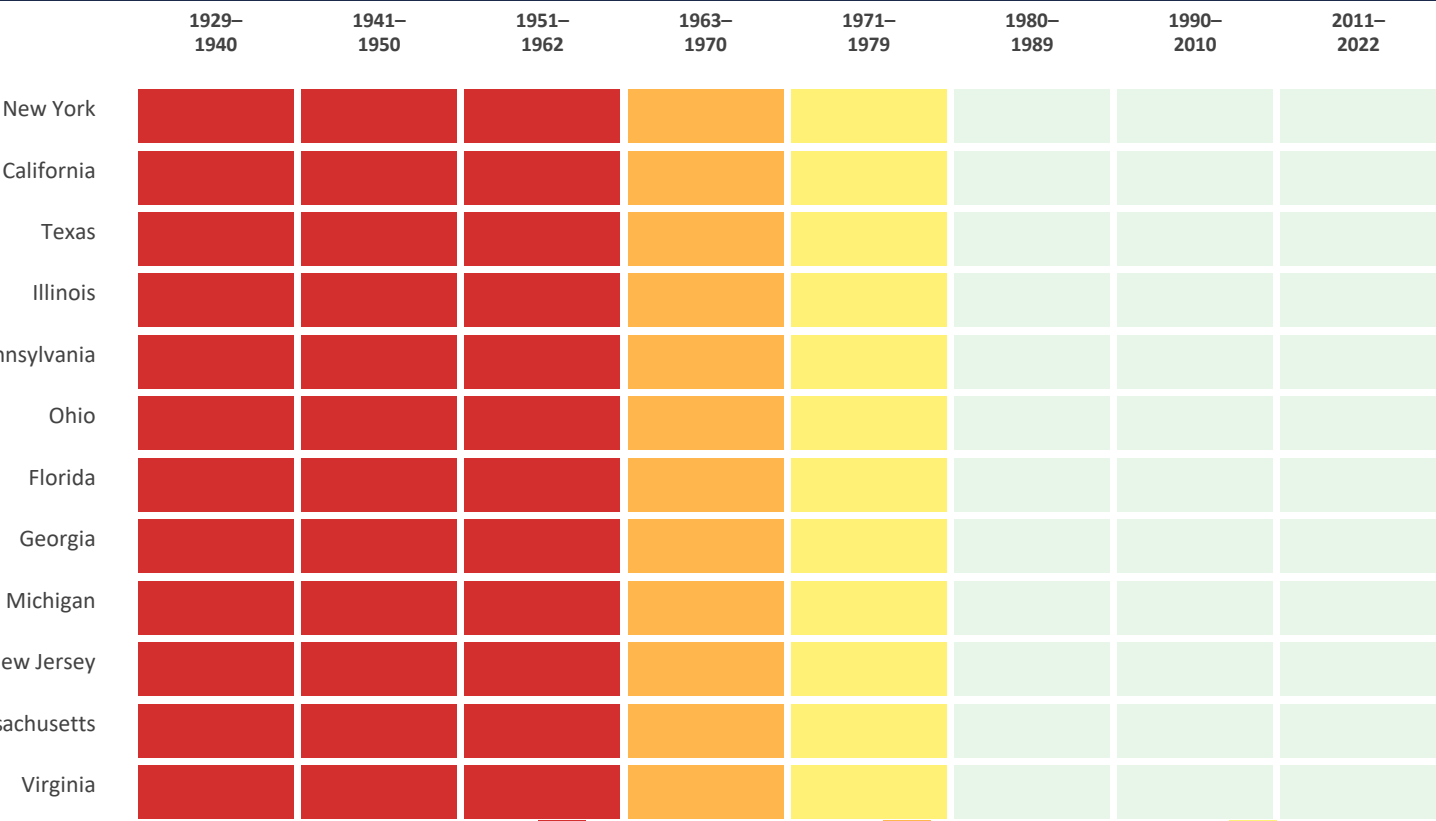
Pre-vaccine vs. post-vaccine annual case and death rates per million population

Disease	Pre-Vaccine Cases/yr	Post-Vaccine Cases/yr	Case Reduction	Pre-Vaccine Deaths/yr	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

■ Green = 100% reduction ■ Red = <90% reduction | Source: Roush & Murphy (2007), via Our World in Data

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

State-by-state case rates per 100,000, 1929–2022



Cases per 100,000: 1,000+ 100–1,000 10–100 Near 0
Pattern consistent across all 50 states — high incidence before 1963, near zero after 1980 mandate

1963 First measles vaccine (John Enders)

1971 MMR vaccine developed (Maurice Hilleman)

1980 Mandatory vaccination for kindergarten entry

Smallpox: The Only Human Disease Eradicated Through Vaccination

From a global scourge to certified eradication in two decades



Why Smallpox Was Uniquely Suited for Eradication

- Virus infected only humans — no animal reservoir
- Symptoms were visually distinguishable
- Vaccine was cheap, stable, and highly effective
- Global political will aligned behind a single goal

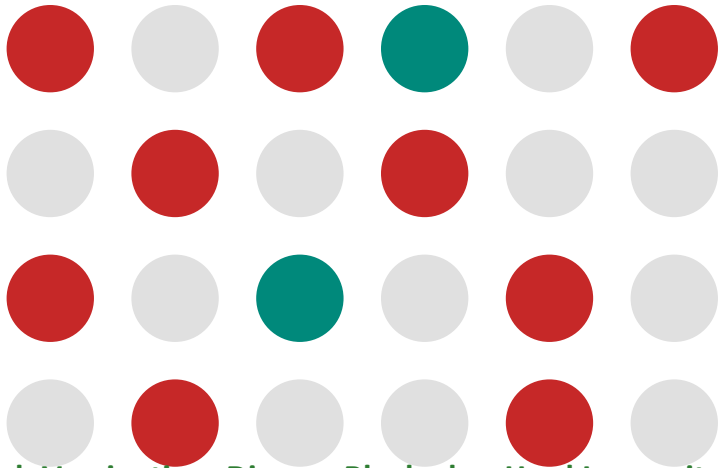
The first vaccine (1796)

Edward Jenner created the first vaccine against smallpox using cowpox to trigger immunity. Nearly two centuries later, that same vaccine technology achieved the only complete eradication of a human disease.

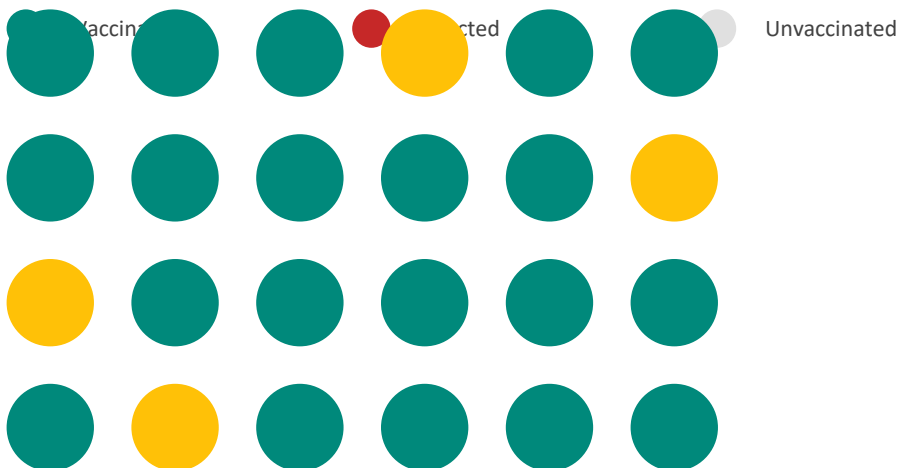
Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, pathogens cannot spread effectively

Low Vaccination: Disease Spreads Freely



High Vaccination: Disease Blocked — Herd Immunity Achieved



How Herd Immunity Works

When a high proportion of a population is vaccinated, pathogens cannot find enough susceptible hosts to sustain transmission chains.

This shields those who cannot be vaccinated:

Newborns — too young for most vaccines

Immunocompromised individuals — vaccines may be contraindicated

Cancer patients on chemotherapy — immune systems suppressed

The threshold varies by disease. Highly contagious diseases like measles require ~95% coverage; less transmissible diseases need lower thresholds.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Not all vaccine-preventable diseases are under control

Pneumococcal Disease

40.5%

case reduction

31.3%

death reduction

233 → 139 cases/million/yr



Hepatitis A

89%

case reduction

88.7%

death reduction

465 → 51 cases/million/yr



Hepatitis B

83.9%

case reduction

83.6%

death reduction

273 → 44 cases/million/yr



Key Barriers to Full Vaccine Impact

Access Gaps

Globally, not everyone has access to vaccines. Low-income countries face infrastructure and cost barriers that limit reach.

Supply Shortages

Vaccine supply chains are fragile. Shortages put children at risk and delay campaigns, especially for newer vaccines.

Trust Challenges

Poor scientific understanding and misinformation can reduce trust and uptake, undermining herd immunity even where vaccines are available.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

Summary of findings and forward-looking priorities

1

100% Elimination Achieved

Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US.

2

Development Timelines Collapsed

Vaccine development timelines have collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.

3

Measles Near Zero Nationwide

Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate.

4

Only One Disease Eradicated

Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global campaign.

5

Coverage Gaps Persist

Gaps persist for pneumococcal disease (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%), highlighting the need for improved access, coverage, and newer vaccines.

Call to Action: Improve Access, Coverage, and Trust in Vaccination Worldwide

The data is clear — vaccines work. The remaining challenge is ensuring they reach everyone who needs them.